

Event-Dependent Bias in Low-Cost PM Sensors for Indoor Environmental Monitoring

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Low-cost particulate matter (PM) sensors increasingly supply indoor air-quality data to smart-building platforms, citizen-science portals, and environmental information systems, where static correction models are commonly applied. This study tests whether such corrections remain valid across distinct indoor aerosol regimes.

During a four-day co-location campaign in a single office (February 2026), AirVisual Pro, Atmotube Pro, Alphasense OPC-N3, and two display-only consumer monitors operated alongside a TSI DustTrak II 8532 photometer and a Palas Fidas Smart 100 aerosol spectrometer (EN 16450 and MCERTS certified; 64 size channels, 32 per decade, 0.18–17.8 μm ; one-minute resolution). Six controlled-source events included ultrasonic diffusion of tap and distilled water, sparkler combustion, cigarette smoking, and incense burning.

Because two monitors lacked data export, their displays were digitised using a fixed camera and a vision-language-model ensemble. Exact-match accuracy increased from 74.3% to 97.2%, mean absolute error decreased from 14.9 to 1.1 $\mu\text{g}/\text{m}^3$ against manually labelled ground truth, and 37.9% of frames were flagged for human review.

Sensor bias was strongly source-dependent and changed sign between events: low-cost devices over-read during humidification but under-read during sparkler combustion. Using the same diffuser, the integrated dose was approximately five to six times higher for tap water than for distilled water, supporting a major contribution from dissolved-solid residue rather than humidity alone. The Fidas conditions its sample using regulated heating, whereas the low-cost devices do not; consequently, humidification events are observed under different aerosol-conditioning states.

Ongoing analyses examine decay time constants to distinguish calibration bias from response dynamics, quantify uncertainty in dose ratios using bootstrap methods, and assess the effects of density and refractive-index assumptions in optical mass conversion. These analyses will inform recommendations for regime-aware calibration, quality flags, and aerosol-conditioning metadata in environmental data infrastructures.

Keywords: low-cost sensors; indoor air quality; data quality; sensor calibration; particulate matter; vision-language models; optical particle sizing

Sustainable Development Goals: SDG 3 (Good Health and Well-being); SDG 11 (Sustainable Cities and Communities)